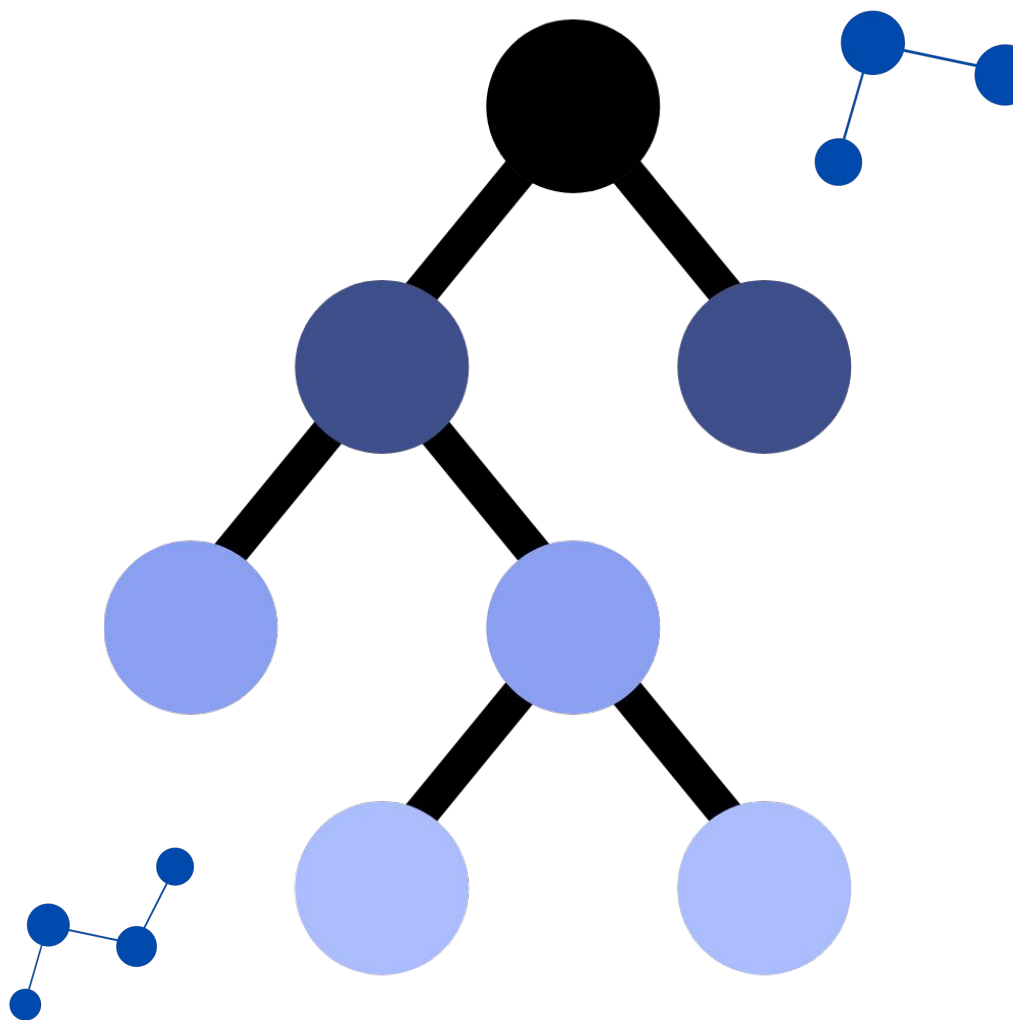


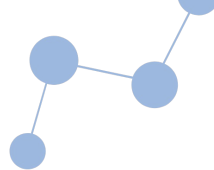
Trees

Binary Trees and Binary
Search Trees

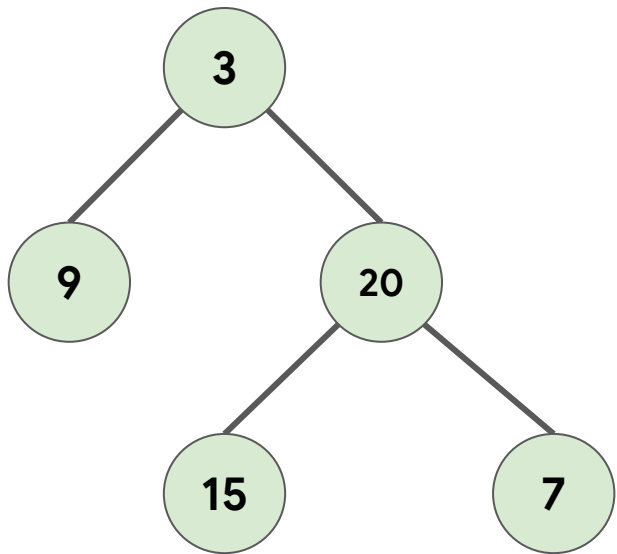
Part II



Simulation



Given the root of a binary tree, return its maximum depth. A binary tree maximum depth is the number of nodes along the longest path from the root node down to the farthest leaf node.



Example 1:

Input: root = [3,9,20,null,null,15,7]

Output: 3

Example 2:

Input: root = [1,null,2]

Output: 2

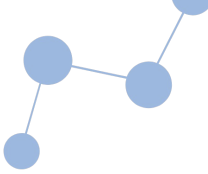
Simulation - Solution



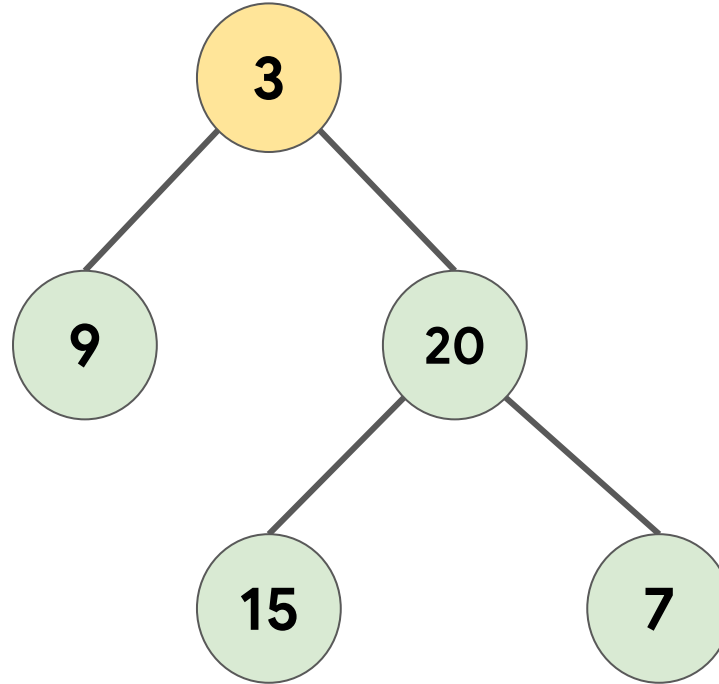
- What if the tree is empty?
Answer: `max_depth = 0`
- What if we have just a node with no children?
Answer: `max_depth = 1`
- What if the current node has left and/or right children?
Answer: `max_depth = 1 + max(left_child_max_depth, right_child_max_depth)`

By recursively calculating the `max_depth` of each subtree

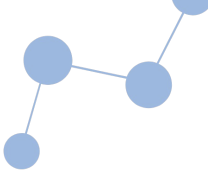
Simulation - Solution



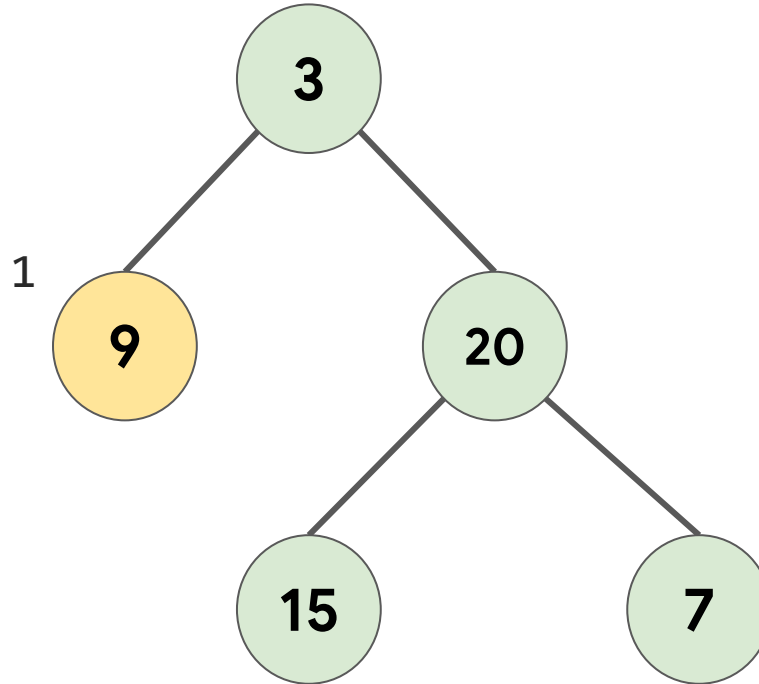
$1 + \text{max}(\text{max_depth}(\text{root.left}), \text{max_depth}(\text{root.right}))$



Simulation - Solution

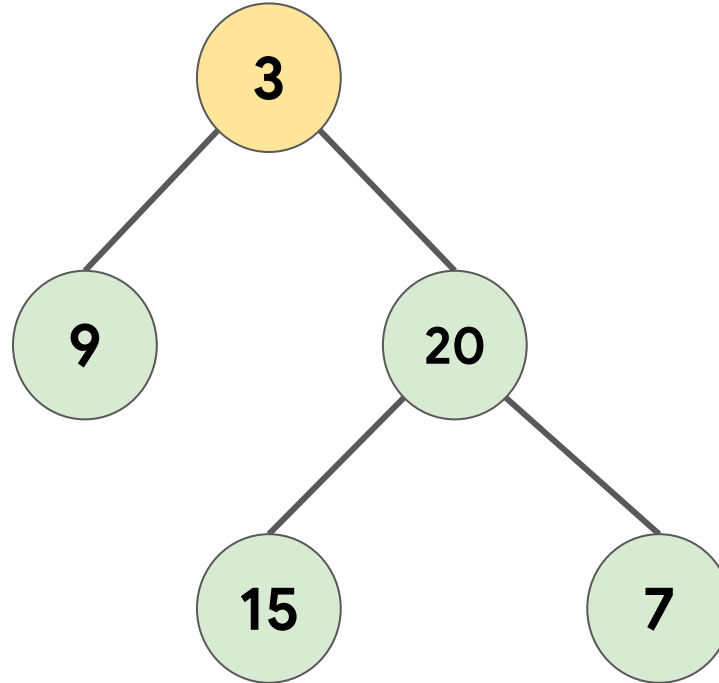


$1 + \text{max}(\text{max_depth}(\text{root.left}), \text{max_depth}(\text{root.right}))$

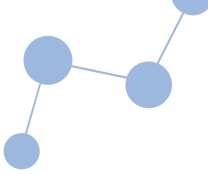


Simulation - Solution

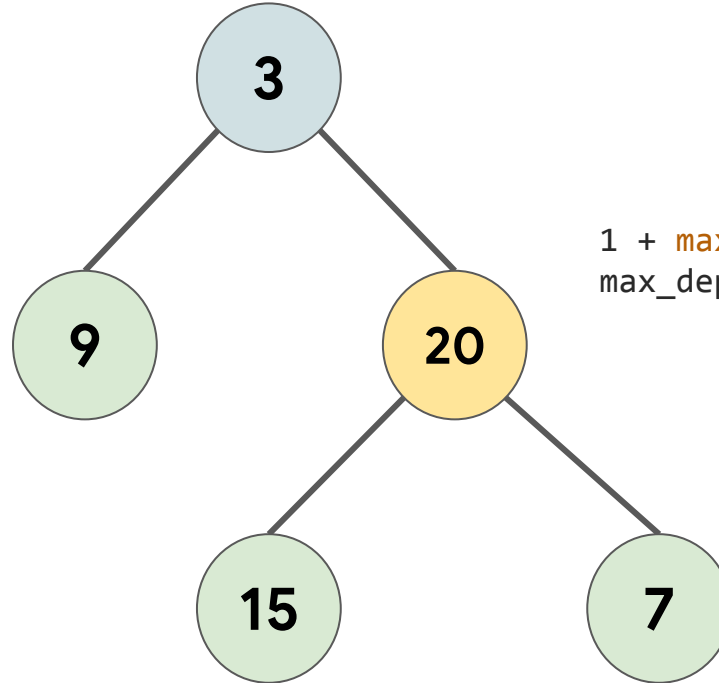
```
1 + max(1, max_depth(root.right))
```



Simulation - Solution

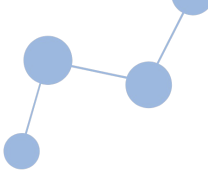


`1 + max(1, max_depth(root.right))`

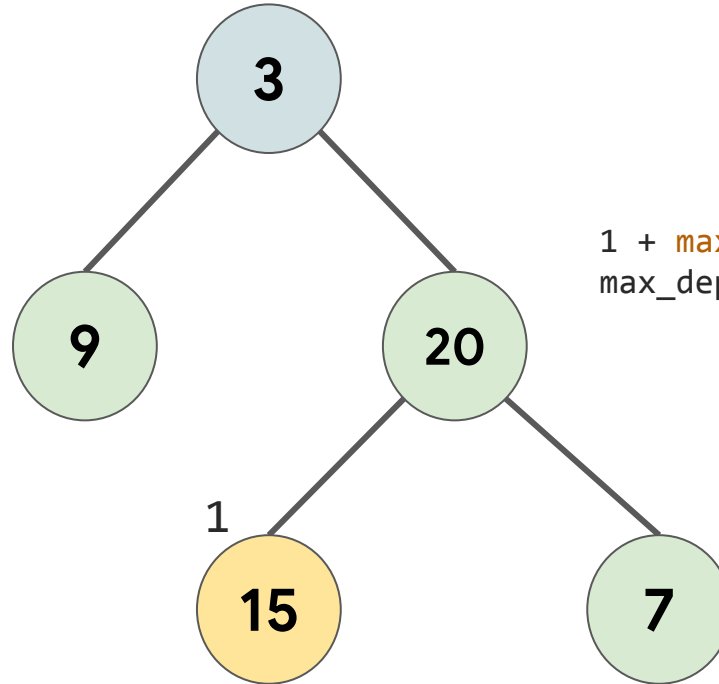


`1 + max(max_depth(root.left),
max_depth(root.right))`

Simulation - Solution

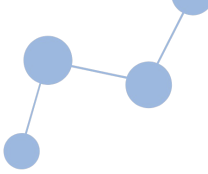


$1 + \text{max}(1, \text{max_depth}(\text{root.right}))$

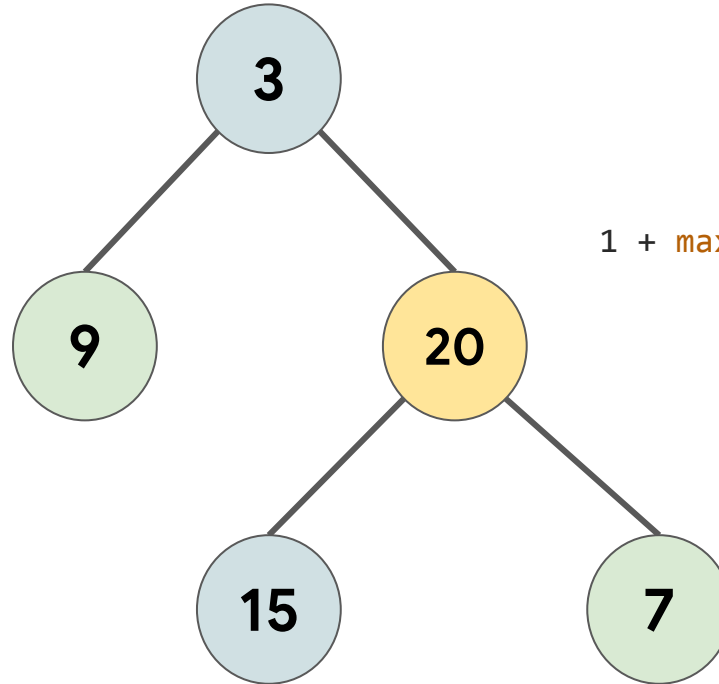


$1 + \text{max}(\text{max_depth}(\text{root.left}), \text{max_depth}(\text{root.right}))$

Simulation - Solution

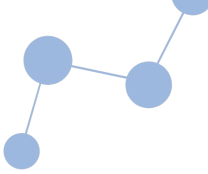


`1 + max(1, max_depth(root.right))`

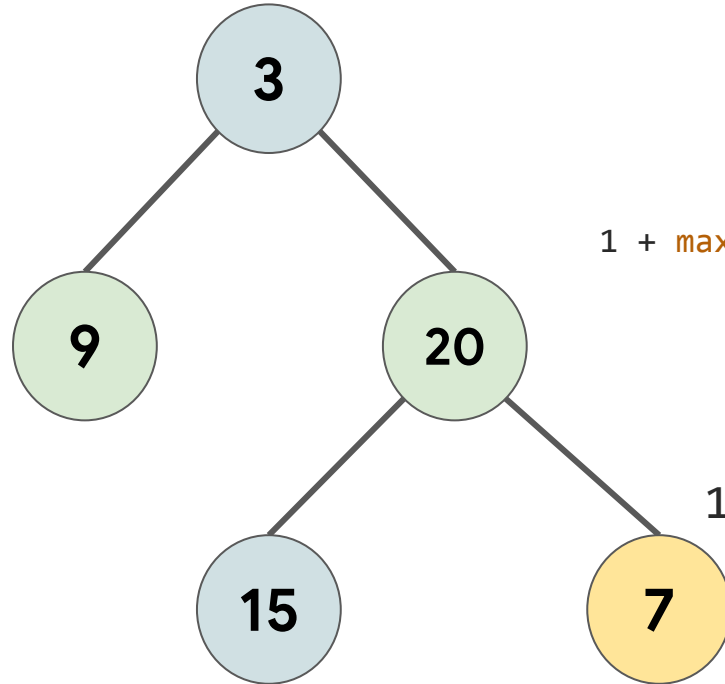


`1 + max(1, max_depth(root.right))`

Simulation - Solution

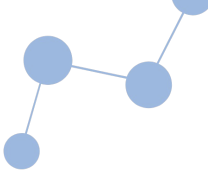


`1 + max(1, max_depth(root.right))`

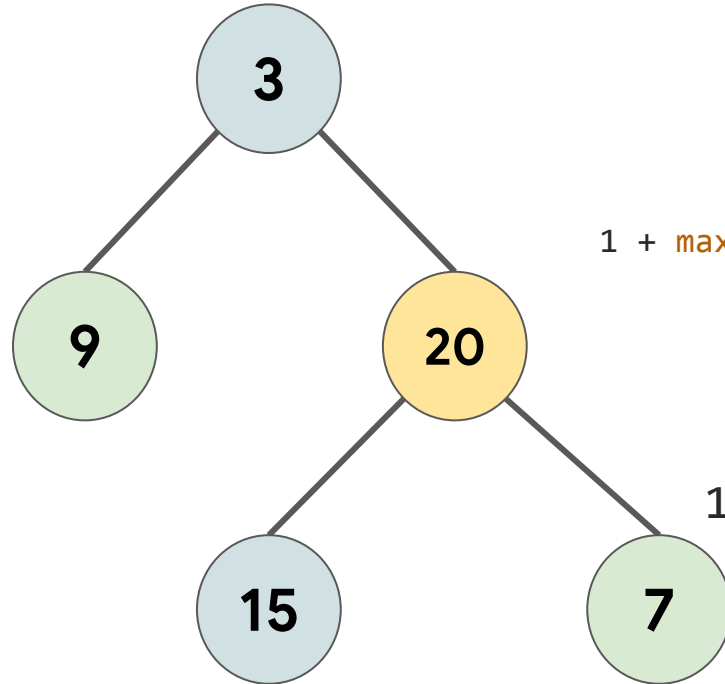


`1 + max(1, max_depth(root.right))`

Simulation - Solution



$1 + \text{max}(1, \text{max_depth}(\text{root.right}))$

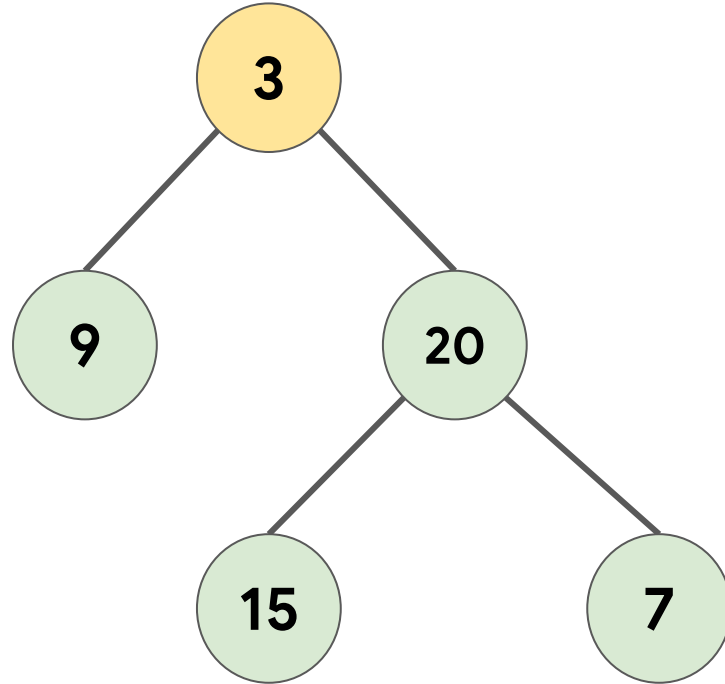


$1 + \text{max}(1, 1) = 2$

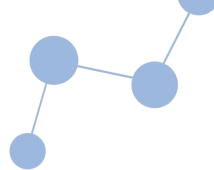
1

Simulation - Solution

$$1 + \text{max}(1,2) = 3$$



Implementation - Recursive

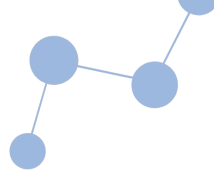


```
# Definition for a binary tree node.
# class TreeNode(object):
#     def __init__(self, val=0, left=None, right=None):
#         self.val = val
#         self.left = left
#         self.right = right
```

```
class Solution:
    def maxDepth(self, root:TreeNode):
        """
        :type root: TreeNode
        :rtype: int
        """
        def find_max(node):
            if not node : return 0
            left = 1 + find_max(node.left)
            right = 1 + find_max(node.right)
            return max(left,right)
        return find_max(root)
```

Time complexity: $O(n)$
Space Complexity: $O(d)$

Implementation - Iterative

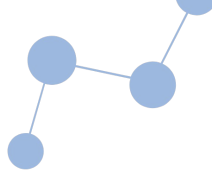


```
class Solution:
    def maxDepth(self, root:TreeNode):
        """
        :type root: TreeNode
        :rtype: int
        """
        if root is None : return 0

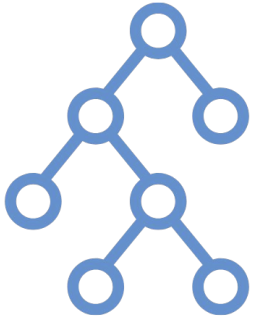
        stack = []
        stack.append((root, 1))
        res = 0

        while stack:
            node, depth = stack.pop()
            if node:
                res = max(res, depth)
                stack.append((node.left, depth+1))
                stack.append((node.right, depth+1))
        return res
```

Time complexity: $O(n)$
Space Complexity: $O(d)$



Basic Operation on Trees

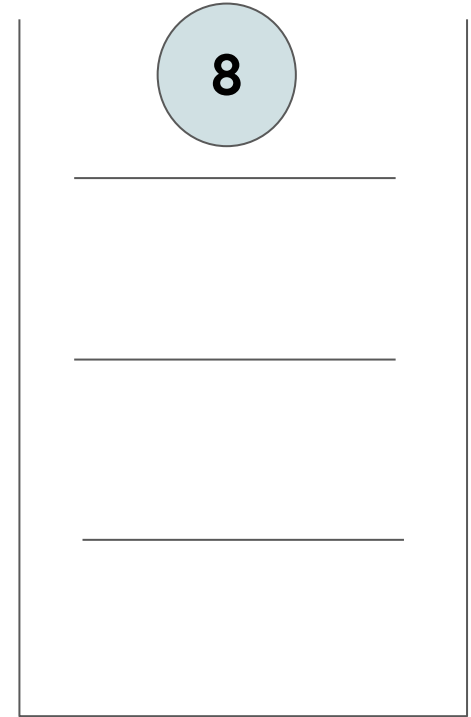
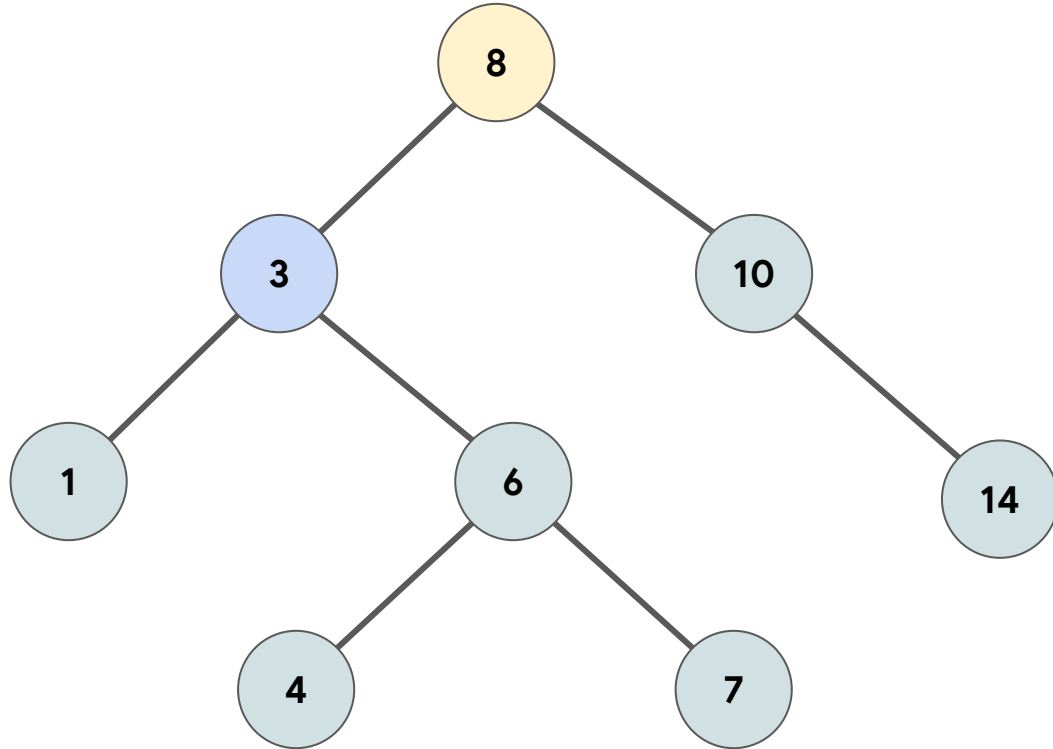


Operation - Searching

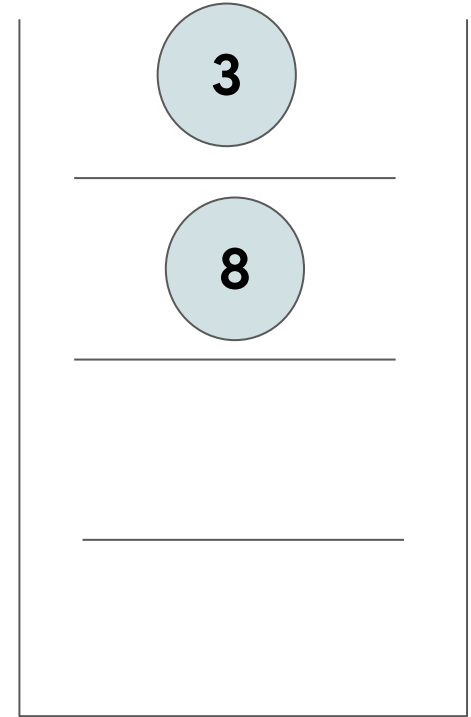
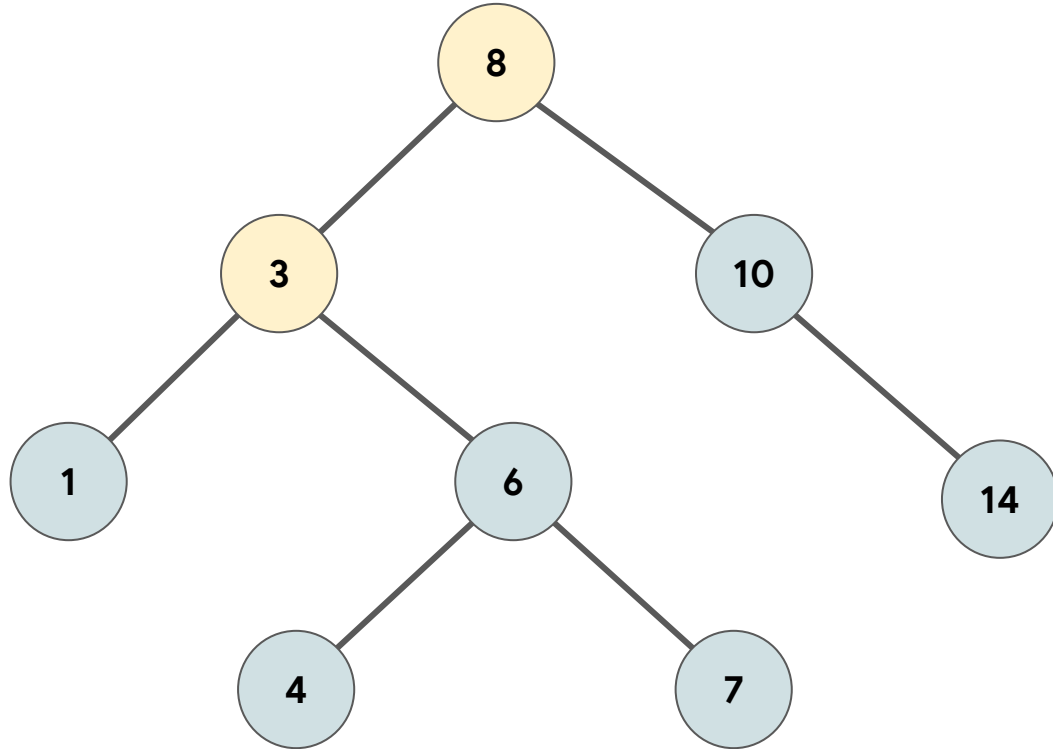


- The algorithm depends on the property of BST that if each left subtree has values below parent and each right subtree has values above the parent.
- If the value is below the parent, we can say for sure that the value is not in the right subtree; we need to only search in the left subtree
- If the value is above the parent, we can say for sure that the value is not in the left subtree; we need to only search in the right subtree.
- Let us try to visualize this with a diagram searching for 4 in the tree:

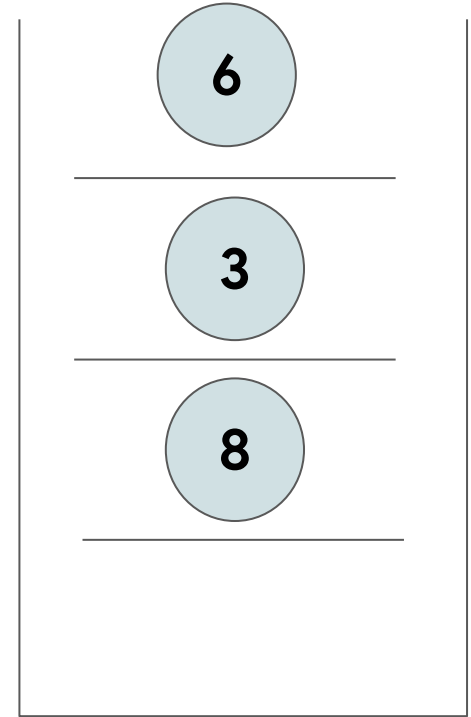
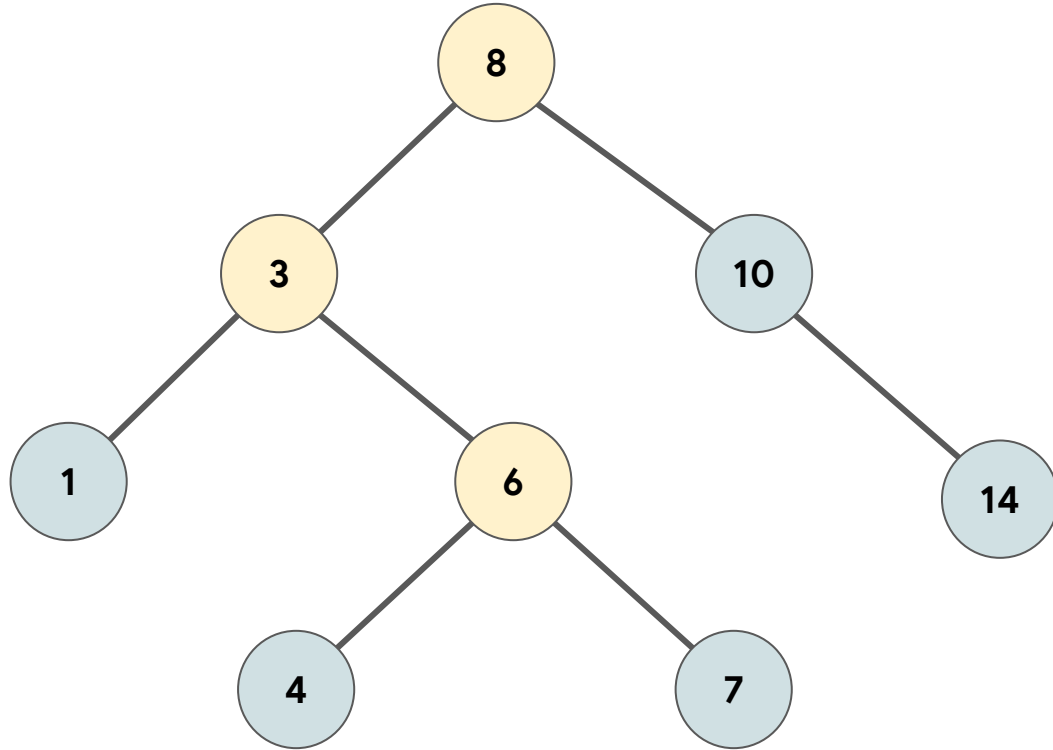
Operation - Searching



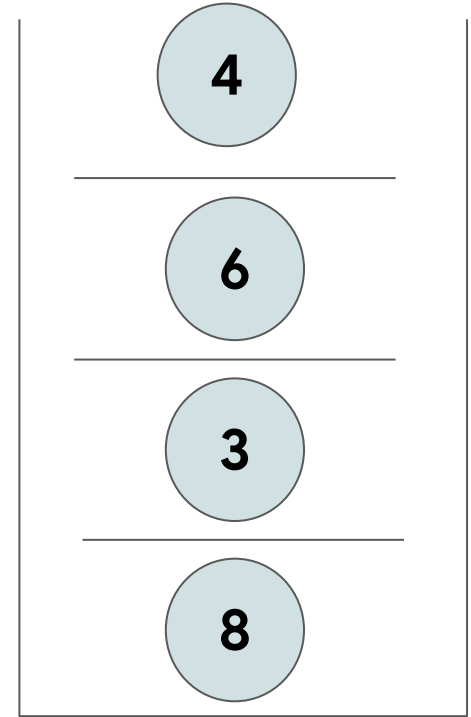
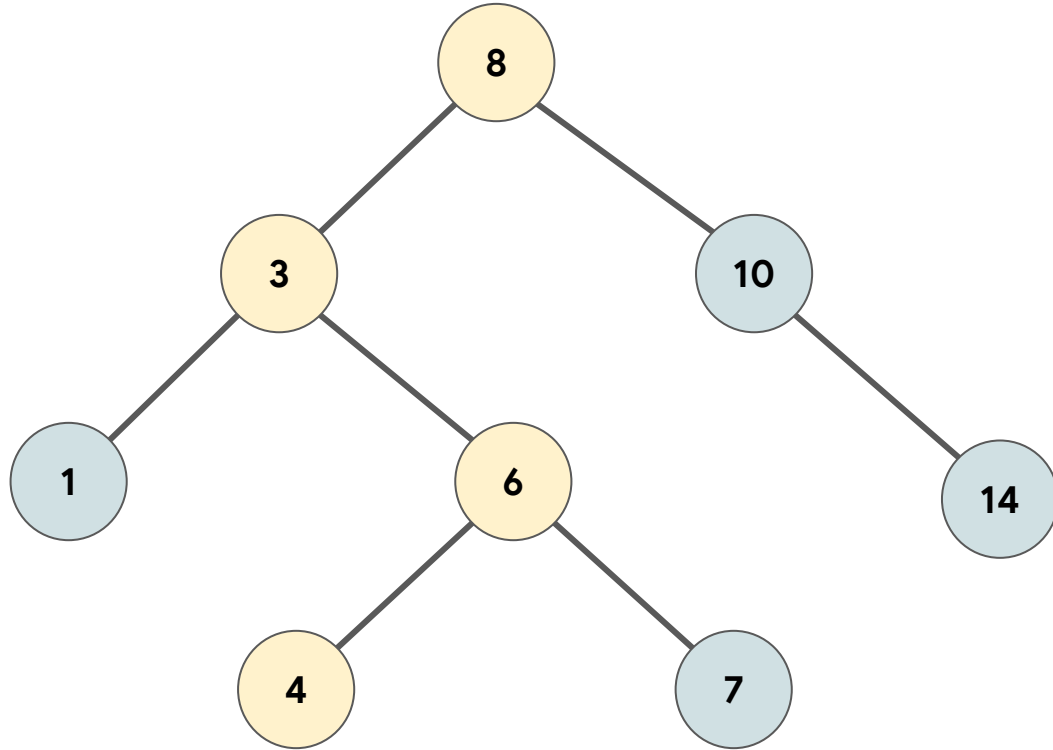
Operation - Searching

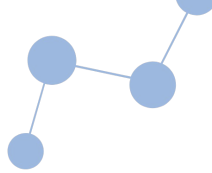


Operation - Searching



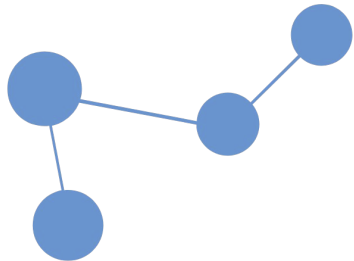
Operation - Searching





Question

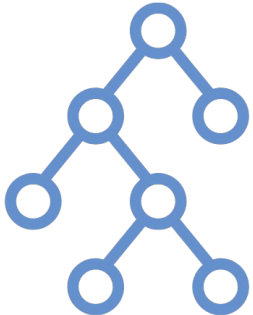
Search in a Binary
Search Tree



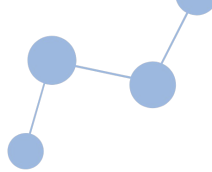
Operation - Searching Algorithm



```
def search(root):  
    if root is None:  
        return None  
    if number == root.val:  
        return root.val  
    if number < root.val:  
        return search(root.left)  
    if number > root.val:  
        return search(root.right)
```



Operation - Insertion

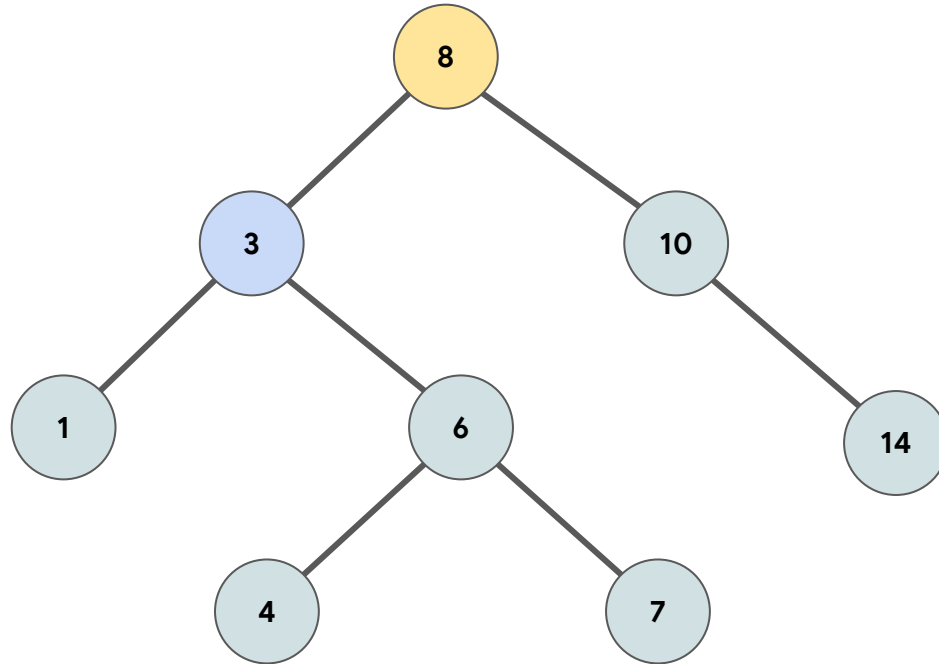


- Inserting a value in the correct position is similar to searching because we **try to maintain the BST rule** that the left subtree is lesser than root and the right subtree is larger than root.
- We keep going to either right subtree or left subtree depending on the value and when we reach a point left or right subtree is null, we put the new node there.

Let's try to visualize how we add a number 5 to an existing BST.

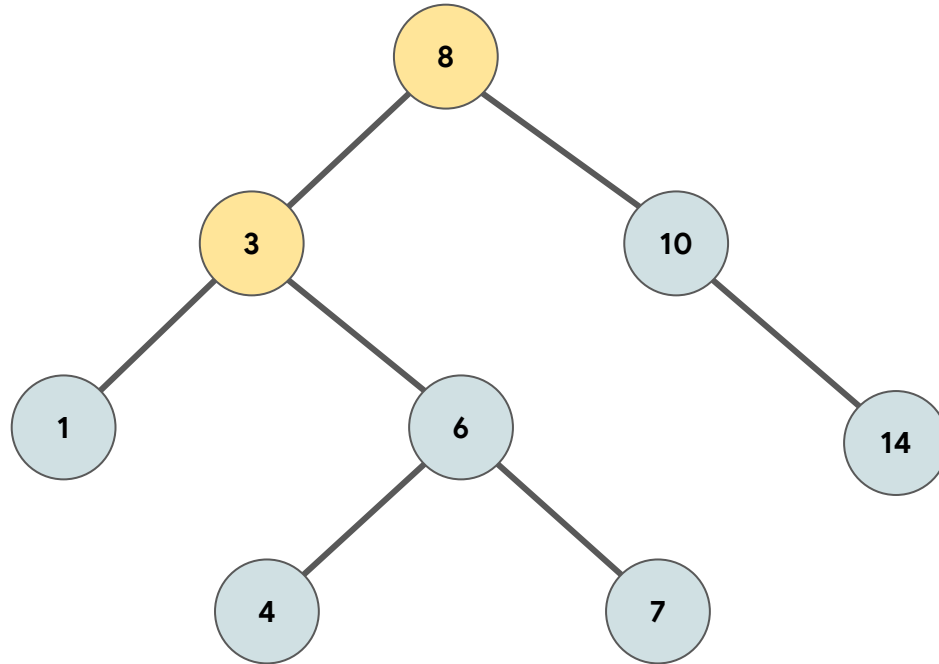
Operation - Insertion

Insert 5 in to the BST



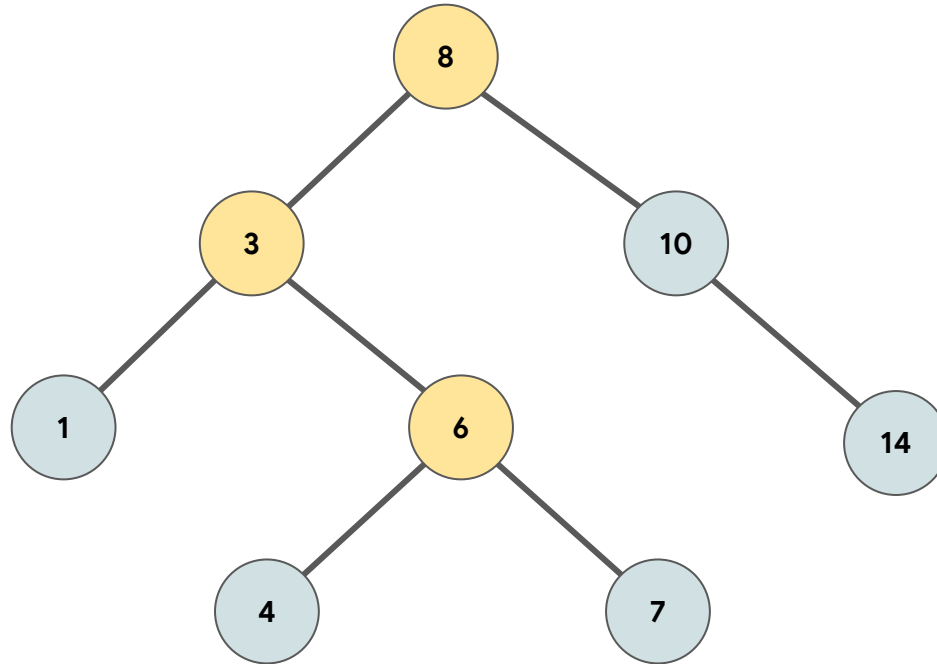
Operation - Insertion

Insert 5 in to the BST



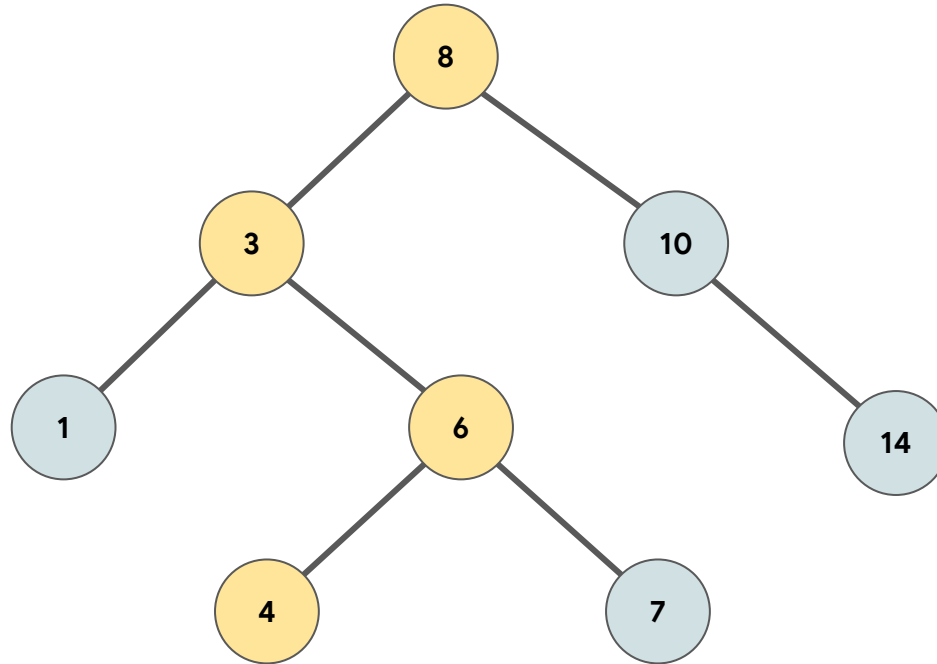
Operation - Insertion

Insert 5 in to the BST



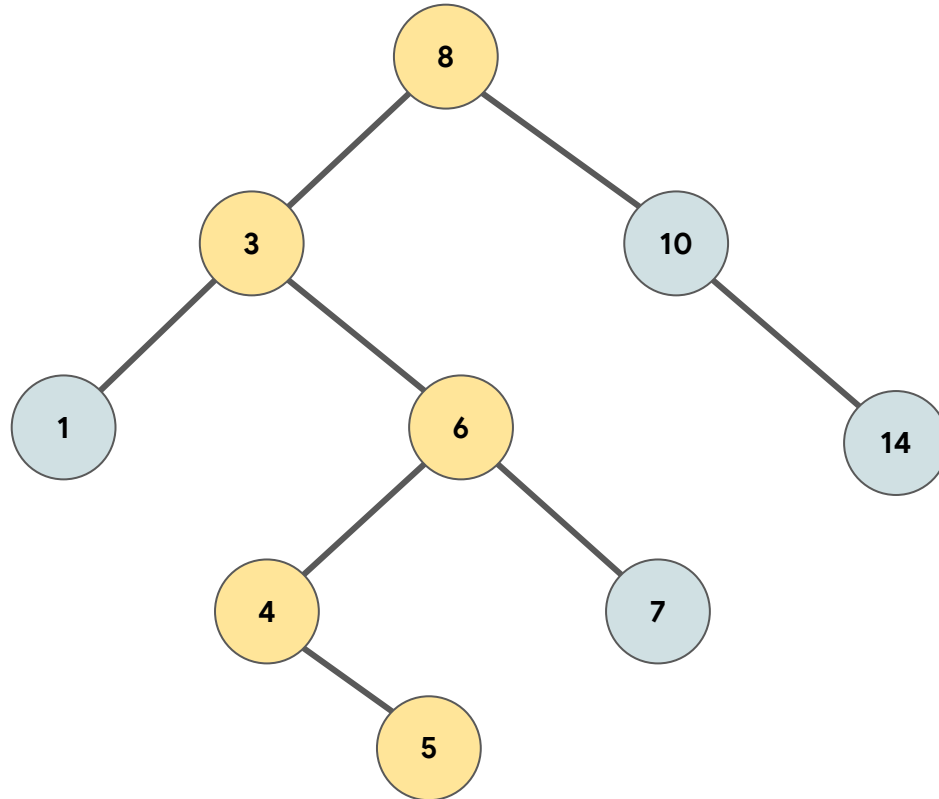
Operation - Insertion

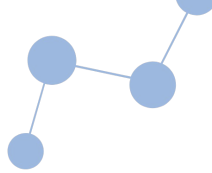
Insert 5 in to the BST



Operation - Insertion

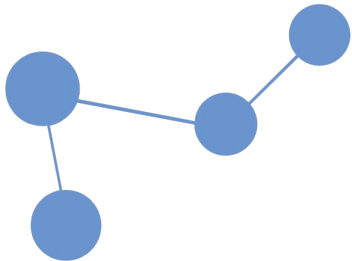
Insert 5 in to the BST



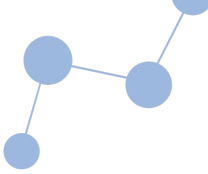


Question

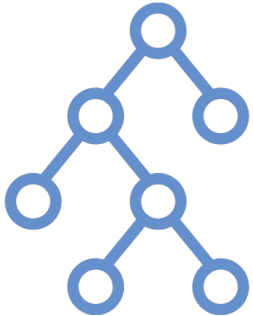
Insert into a Binary
Search Tree



Operation - Insertion Algorithm



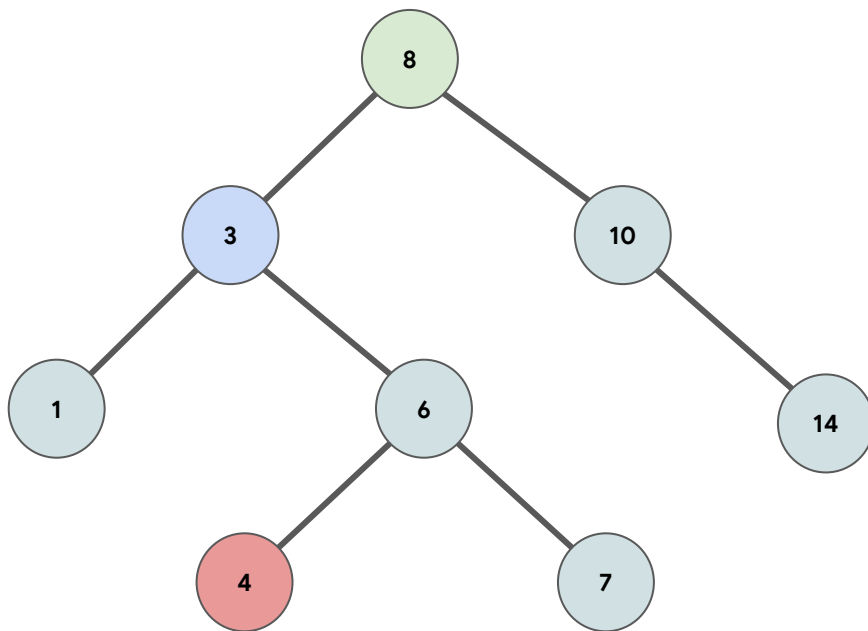
```
def insert(root, data):  
    if root is None:  
        return Node(data)  
    if data < root.val:  
        root.left = insert(root.left, data)  
        return root  
    if data > root.val:  
        root.right = insert(root.right, data)  
        return root
```



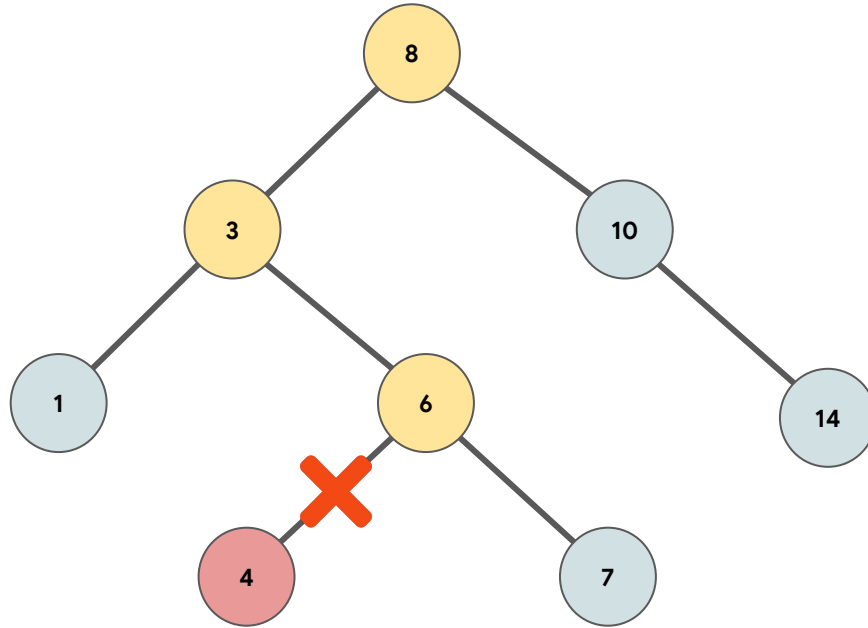
Operation - Deletion

- There are three cases for deleting a node from a binary search tree.

Case One: In the first case, the node to be deleted is the leaf node. In such a case, simply delete the node from the tree. 4 is to be deleted.



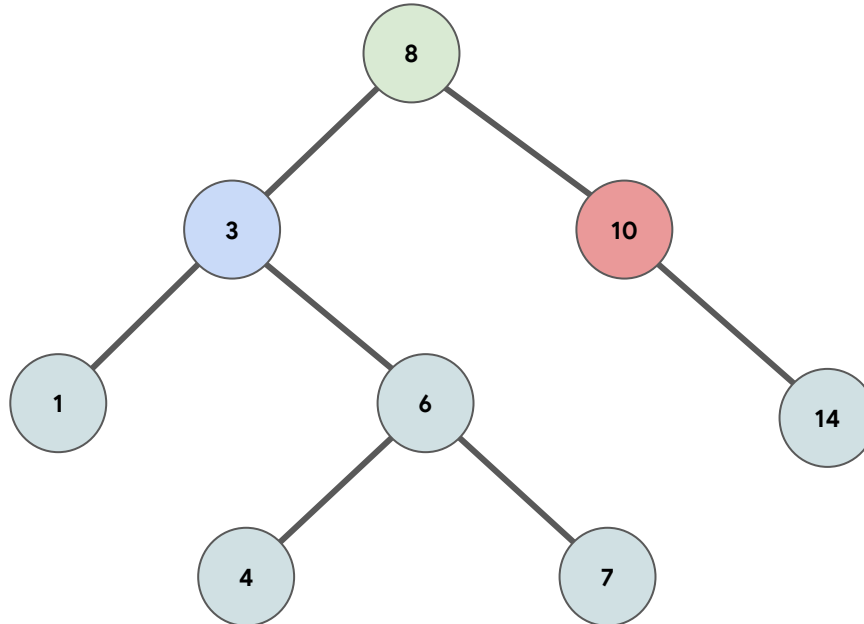
Operation - Deletion



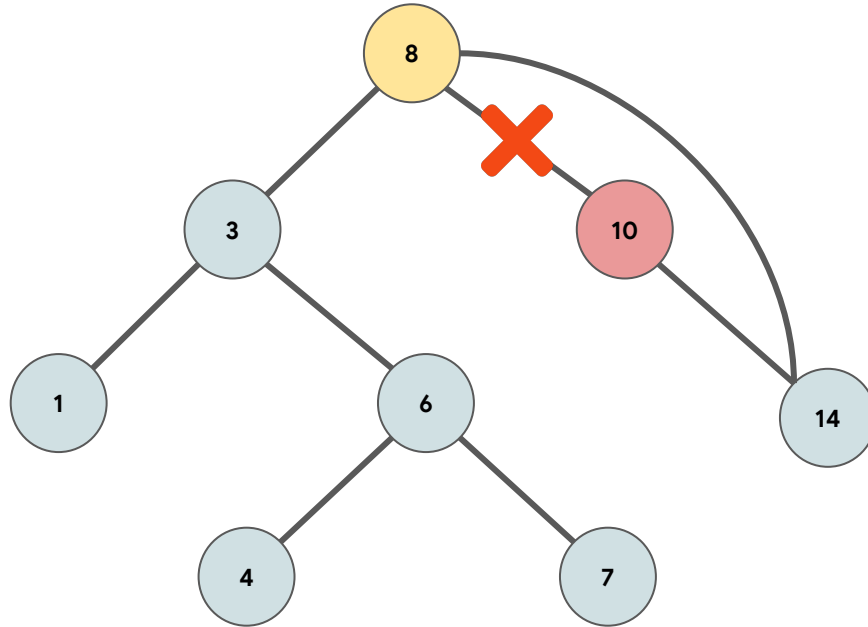
Operation - Deletion

Case Two: In the second case, the node to be deleted lies has a single child node. In such a case follow the steps below:

1. Replace that node with its child node.
2. Remove the child node from its original position.



Operation - Deletion



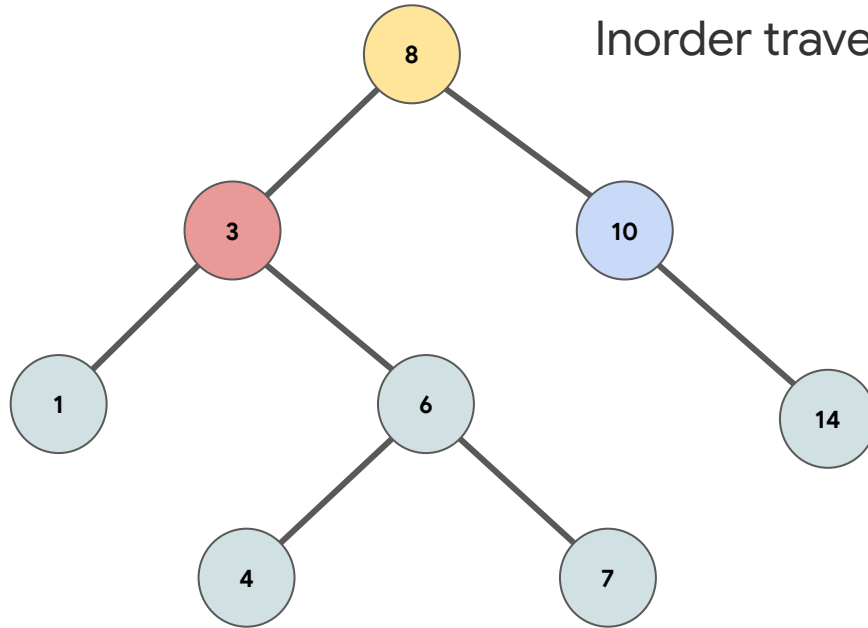
Operation - Deletion



Case Three: The node to be deleted has two children. In such a case follow the steps below:

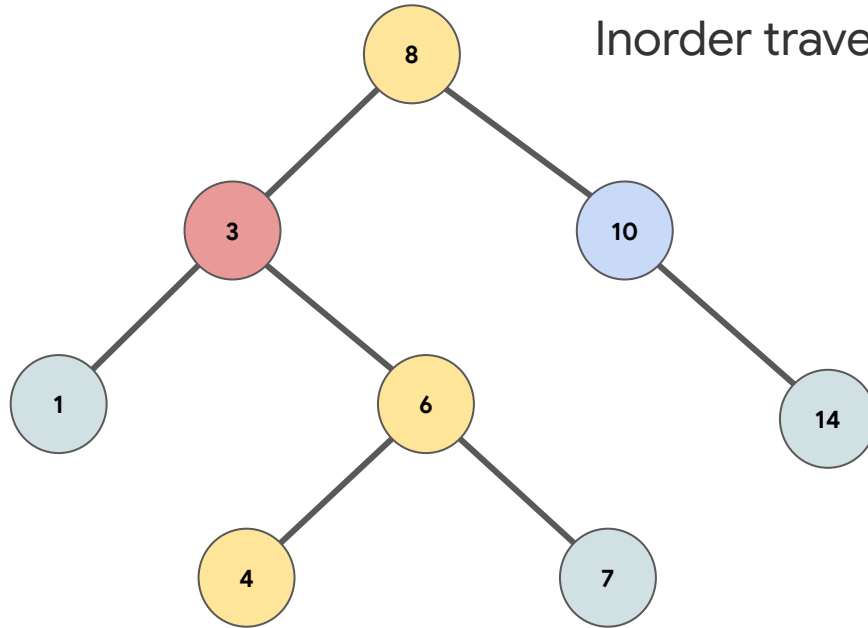
1. Get the **inorder successor** of that node. Why?
2. Replace the node with the inorder successor.
3. Remove the inorder successor from its original position.

Operation - Deletion



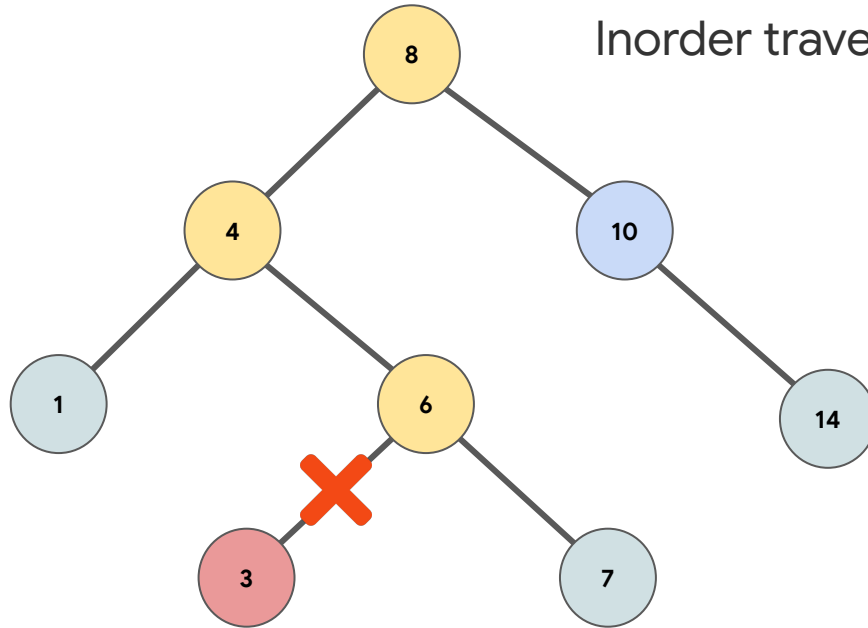
Inorder traversal: 1 3 4 6 7 8 10 14

Operation - Deletion

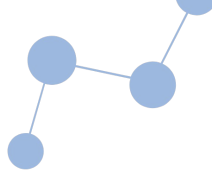


Inorder traversal: 1 3 4 6 7 8 10 14

Operation - Deletion

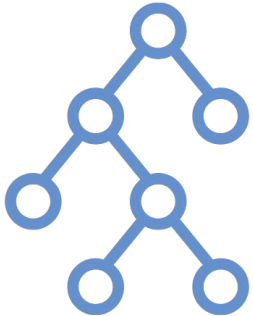


Inorder traversal: 1 4 6 7 8 10 14

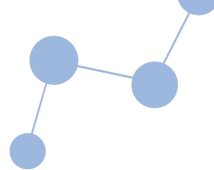


Practice Problem

Delete Node in a BST



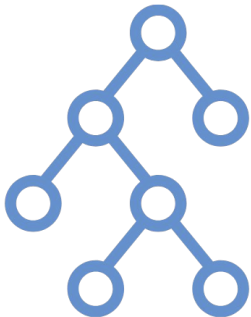
Operation - Deletion Algorithm



```
def deleteNode(root, key):
    # Return if the tree is empty
    if not root:
        return root
    # Find the node to be deleted
    if key < root.key:
        root.left = deleteNode(root.left, key)
    elif(key > root.key):
        root.right = deleteNode(root.right, key)
    else:
        # If the node is with only one child or no child
        if not root.left:
            return root.right
        elif not root.right:
            return root.left

        # If the node has two children,
        # place the inorder successor in position of the node to be deleted
        temp = minValueNode(root.right)
        root.key = temp.key

        # Delete the inorder successor
        root.right = deleteNode(root.right, temp.key)
    return root
```

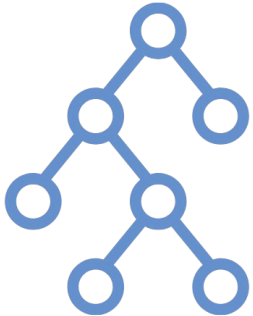


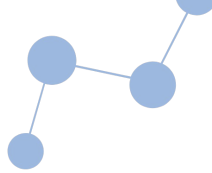
Operation - Deletion Algorithm

```
`# Find the inorder successor
def minValueNode(node):
    current = node

    # Find the leftmost leaf
    while current.left:
        current = current.left

    return current
```





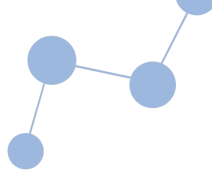
Time and Space Complexity Analysis

Binary Tree

- Traversing
 - Time = ?
- Searching
 - Time = ?
- Insertion
 - Time = ?
- Deletion
 - Time = ?
- Space = ?

Binary Search Tree

- Traversing
 - Time = ?
- Searching
 - Time = ?
- Insertion
 - Time = ?
- Deletion
 - Time = ?
- Space = ?



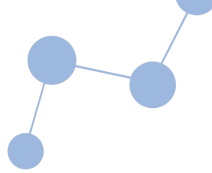
Time and Space Complexity Analysis

Binary Tree

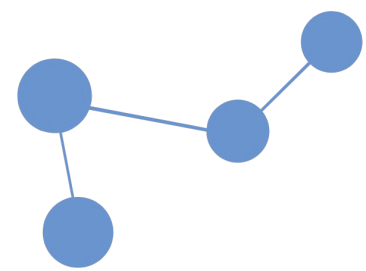
- Traversing
 - Time = $O(n)$
- Searching
 - Time = $O(n)$
- Insertion
 - Time = $O(n)$
- Deletion
 - Time = $O(n)$
- Space = $O(n)$... Why?

Binary Search Tree

- Traversing
 - Time = $O(n)$.
- Searching
 - Time = $O(h)$ where h is the height of BST
- Insertion
 - Time = $O(h)$
- Deletion
 - Time = $O(h)$
- Space = $O(n)$

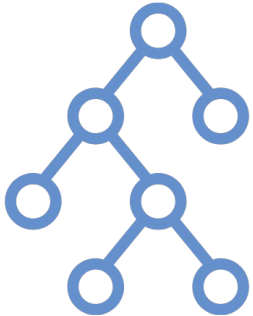


Checkpoint link

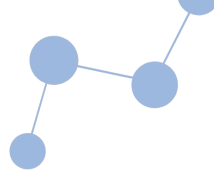


Common Pitfalls

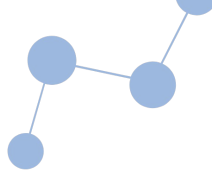
- Null pointer exceptions
- Assuming the tree is balanced
- Wrong choice of traversal
- Wrong recurrence relations and base cases
- Stack overflow



Applications of Trees

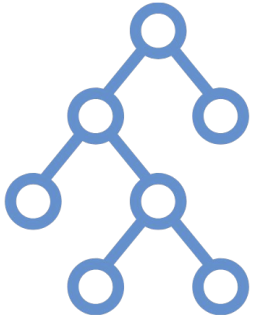


- Representation structure in **File Explorer**. (Folders and Subfolders) uses N-ary Tree.
- **Auto-suggestions** when you google something using Trie.
- Used in **decision-based machine learning algorithms**.
- Tree forms the backbone of other complex data structures like heap, priority queue, spanning tree, etc.
- A binary tree is used in **database indexing** to store and retrieve data in an efficient manner.
- **Binary Search Trees (BST)** can be used in **sorting algorithms**.

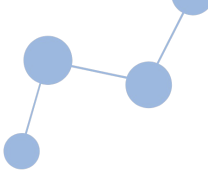


Practice Questions

- [Merge Two Binary Trees](#)
- [Search in Binary Search Trees](#)
- [Same Tree](#)
- [Lowest Common Ancestor of Binary Search Tree](#)
- [Validate Binary Search Trees](#)
- [Binary Tree Zigzag Level Order Traversal](#)
- [Maximum Difference Between Node and Ancestor](#)
- [Kth smallest Element in BST](#)
- [Maximum Sum BST in Binary Tree](#)



Quote of the Day



"A tree with strong roots laughs at storms."
- Malay Proverb

